United Airlines Flight Delays

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Introduction

- We analyzed the nycflights13 dataset, focusing on United Airlines (UA) departure and arrival delays.
- Our project explores how much time UA flights lose or recover during travel.
- Specifically, we investigated:
  - Do flights that depart late make up time in the air?
  - What are the five most common UA destinations from NYC?
  - How does the average gain differ across these destinations?
  - Does gain per hour differ for longer flights vs. shorter flights?
  - Our overall goal was to understand whether United Airlines flights recover delayed time once airborne, and what factors influence this.

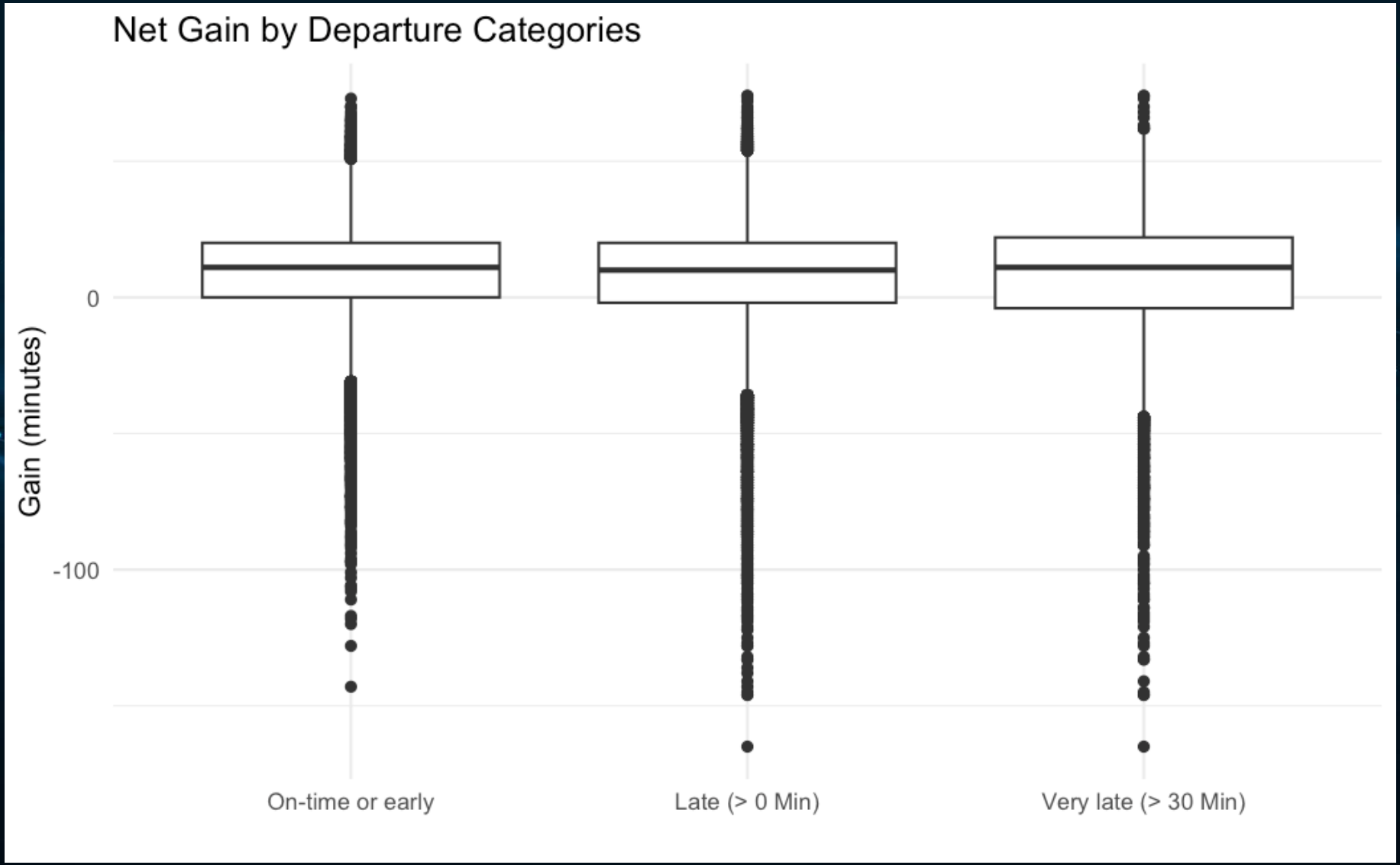
Data Preparation

- Filtered the dataset to include **only United Airlines flights**
- Created new variables (e.g., **time gain**) for analysis
- Cleaned and organized the data for visualizations and statistical tests
- Built charts and graphs to explore trends in delays and gain
- Developed statistical models to better understand patterns in flight recovery time

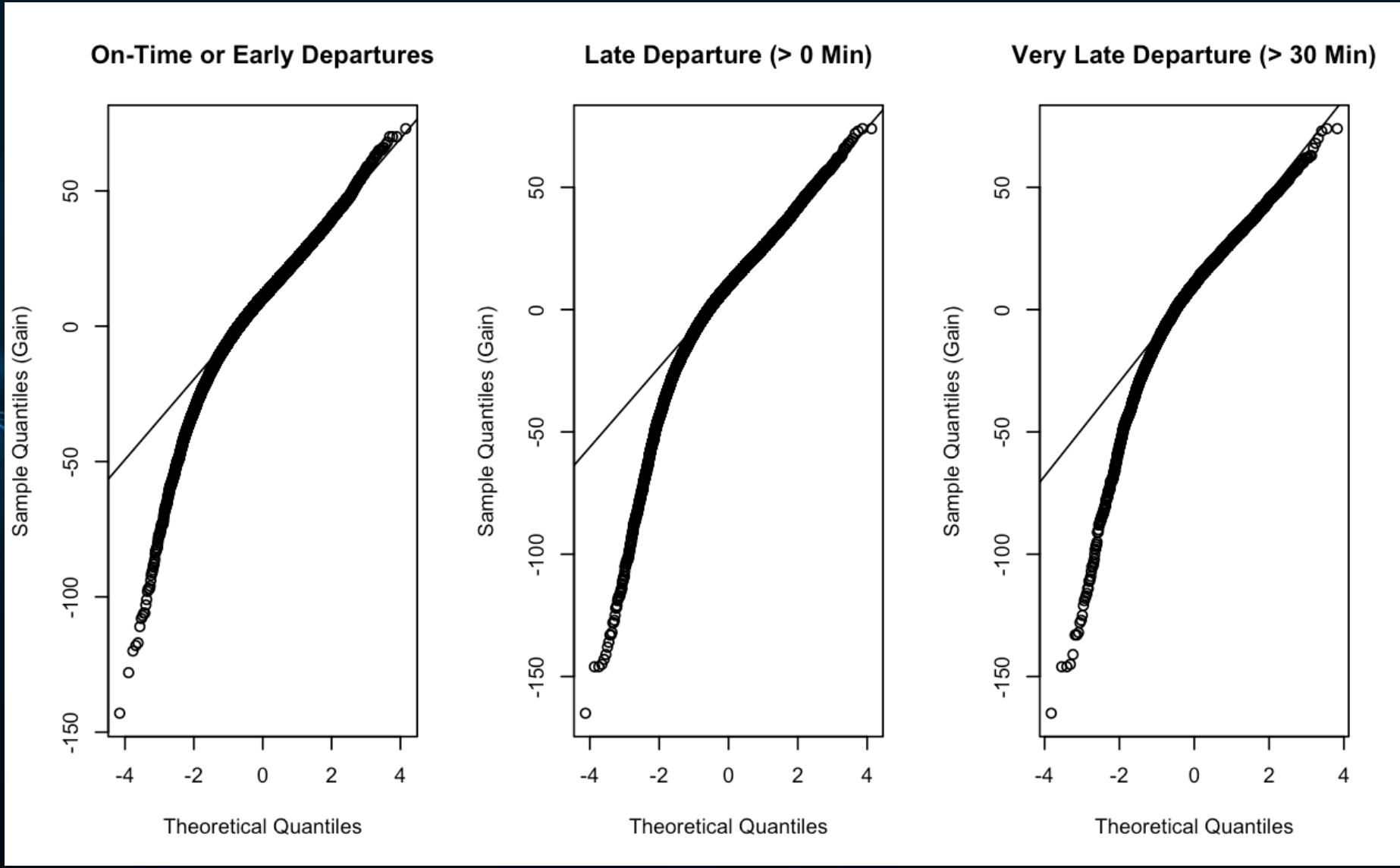
Difference in Average Gain for Ontime/Early, Late, and Very Late Flights

- Compared 3 subgroups of flight gain graphically:
  - Ontime/Early
  - Late (> 0 Minutes)
  - Very Late (> 30 Minutes)
- Conducted Weltch two sample t-tests:
  - Ontime/Early vs Late (> 0 Minutes)
  - Flights <= 30 Minutes vs Very Late (> 30 Minutes)

Distribution of the Data



Further Exploration: Distribution of the Data



Results of Gain by Subgroups, Ontime/Early, Late, Very Late

On-time/early flights gained 1.41 - 2.04 more minutes

Flights that departed 30 min, or less, late gained 1.27 - 2.42 more minutes

Both Welch tests showed p values << 00.1

comparison <chr>	mean_false <dbl>	mean_true <dbl>	lower <dbl>	upper <dbl>	p_value <dbl>
Late	9.269172	7.543115	1.411308	2.040805	2.200e-16
Very Late	8.699534	6.857881	1.268195	2.415112	3.215e-10

What are the five most common destination airports for UA flights from NYC?

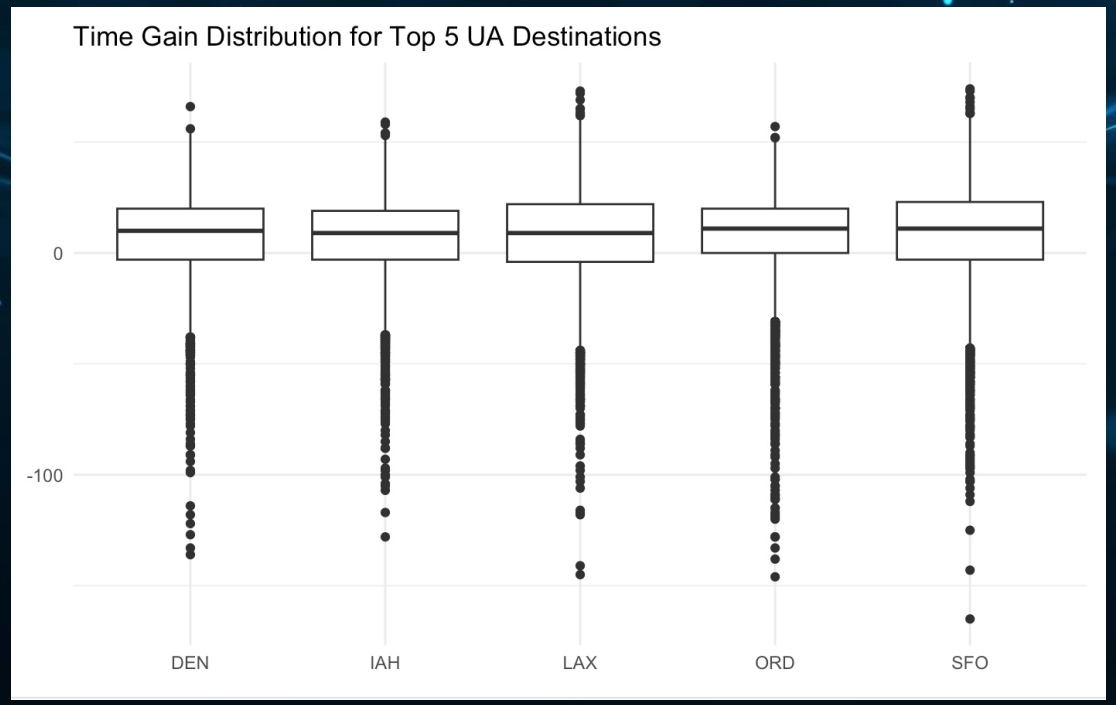
dest <chr>	name <chr>	n <int>
ORD	Chicago Ohare Intl	6984
IAH	George Bush Intercontinental	6924
SFO	San Francisco Intl	6819
LAX	Los Angeles Intl	5823
DEN	Denver Intl	3796

Distribution and Average Gain for Each of the Five Airports

dest <chr>	name <chr>	avg_gain <dbl>	median_gain <dbl>	sd_gain <dbl>	n <int>
ORD	Chicago Ohare Intl	7.777432	11	19.15717	6984
IAH	George Bush Intercontinental	6.861755	9	18.44106	6924
SFO	San Francisco Intl	8.695006	11	22.40789	6819
LAX	Los Angeles Intl	7.825303	9	21.91669	5823
DEN	Denver Intl	7.302382	10	20.04962	3796

- United flights recovered about **7–9 minutes** on average
- Mean gain was similar across all five destinations
- Medians were slightly positive
- High variability shown by large standard deviations

Distribution and Average Gain for Each of the Five Airports



- Similar gain patterns across all five destinations
- Slightly positive medians for each airport
- Wide variability with many outliers
- Average gains fall between **7–9 minutes**

Modeling: Average Gain

```
Residuals:
     Min       1Q   Median       3Q      Max 
-146.583   -7.428    2.305   10.540   49.077 

Coefficients:
              Estimate Std. Error  t value Pr(>|t|)    
(Intercept) 183.500893  17.764754   10.329   <2e-16 ***
destIAH     -21.400433   2.280440   -9.384   <2e-16 ***
destLAX      80.195927   9.378007    8.551   <2e-16 ***
destORD     -86.249674   9.775281   -8.823   <2e-16 ***
destSFO      95.076325  10.602387    8.967   <2e-16 ***
distance     -0.002106   0.011067   -0.190    0.849    
air_time     -0.769298   0.006064 -126.862   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 16.44 on 29786 degrees of freedom
  (553 observations deleted due to missingness)
Multiple R-squared:  0.3533,	Adjusted R-squared:  0.3532 
F-statistic:  2712 on 6 and 29786 DF,  p-value: < 2.2e-16
```

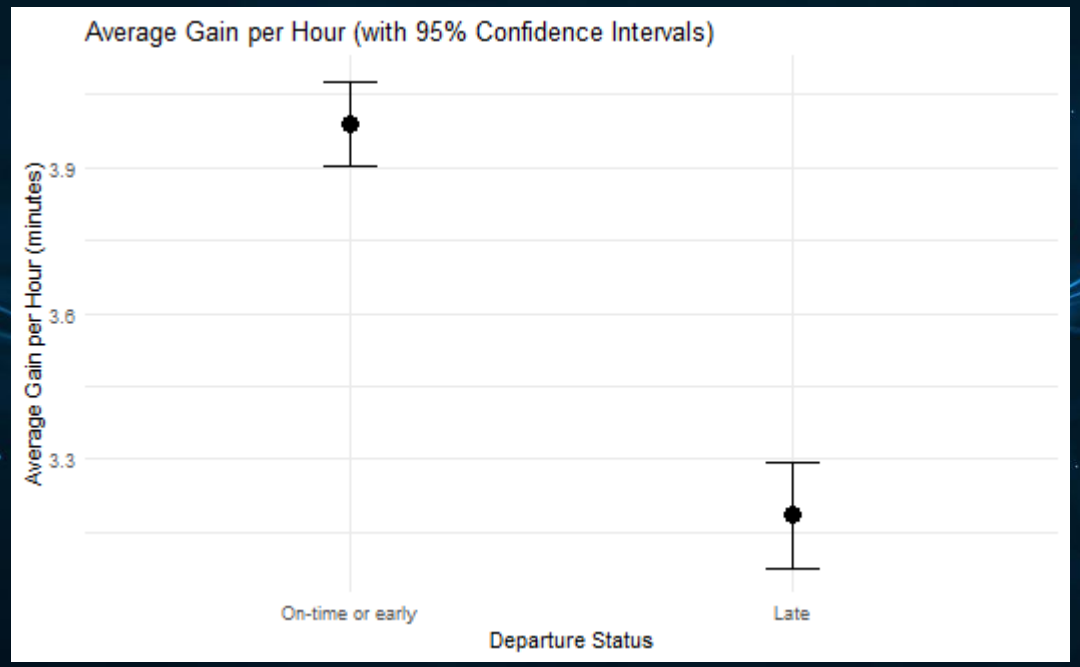
- Used a **multiple linear regression model**
- Air time was the strongest predictor
- Distance was not significant
- Destination effects remained significant

Do Flights That Depart Late Gain More Time in the Air?

- Analyzed United Airlines flights from the nycflights13 dataset
- Created a new variable, gain per hour, to measure time recovered in flight
- Compared gain per hour for on-time vs late departures
- Compared flights delayed >30 minutes vs ≤30 minutes
- Statistical tests: Welch two-sample t-tests and 95% confidence intervals

```{r}
ua <- flights %>%
  filter(carrier == "UA") %>%
  drop_na(dep_delay, arr_delay, air_time) %>%
  mutate(
    net_gain = dep_delay - arr_delay,
    air_time_hr = air_time / 60,
    gain_per_hour = net_gain / air_time_hr,
    late_dep = dep_delay > 0,
    late30_dep = dep_delay > 30
  )
```

Average Gain per Hour for On-time vs Late Departures



95% confidence intervals estimated using Welch two-sample t-test

On-time/early flights gained 3.90 minutes per hour

Late departures gained 3.18 minutes per hour

Difference is statistically significant ($p < .001$)

What This Means for United Airlines?

Late flights do not recover time more effectively during flight.

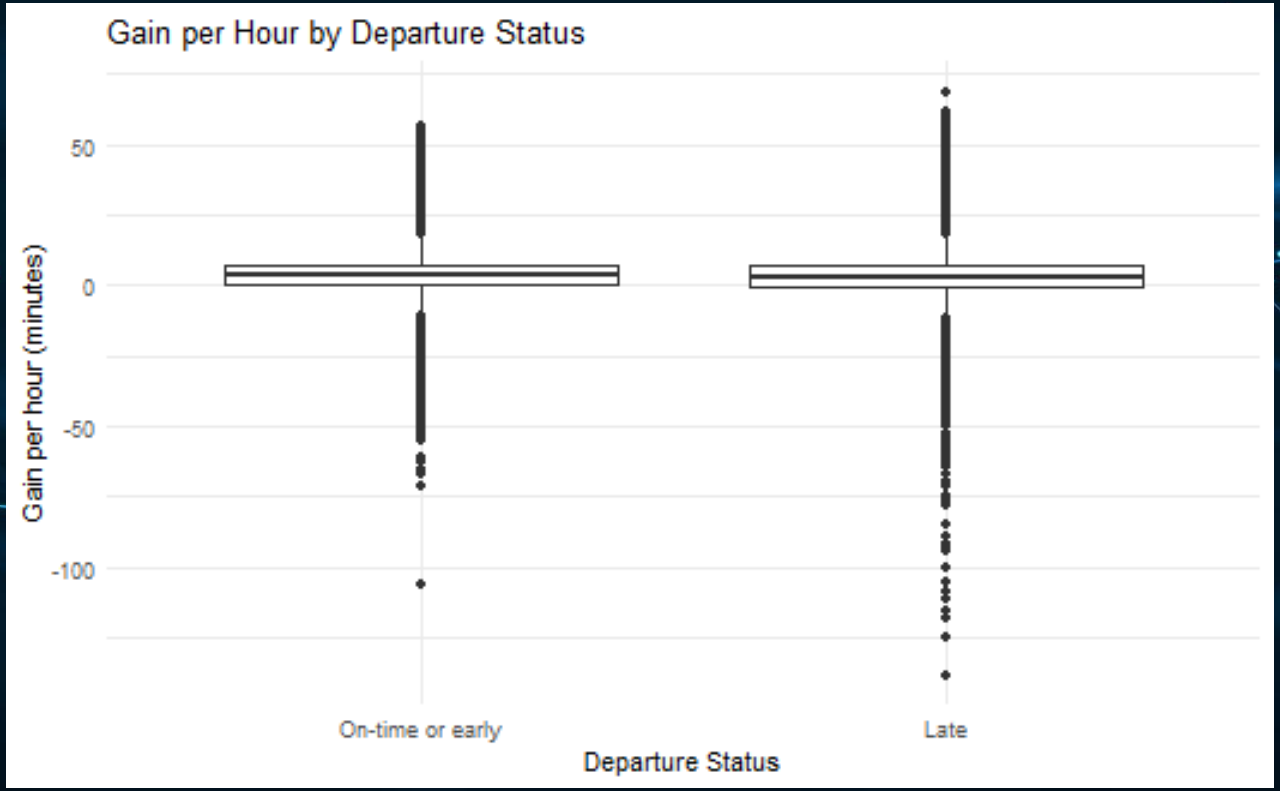
Flights delayed more than 30 minutes gain the least time per hour.

In-air time recovery has limits - ground delays matter more than airborne speed

Improving boarding, gate sequencing, and runway scheduling could reduce future delays.

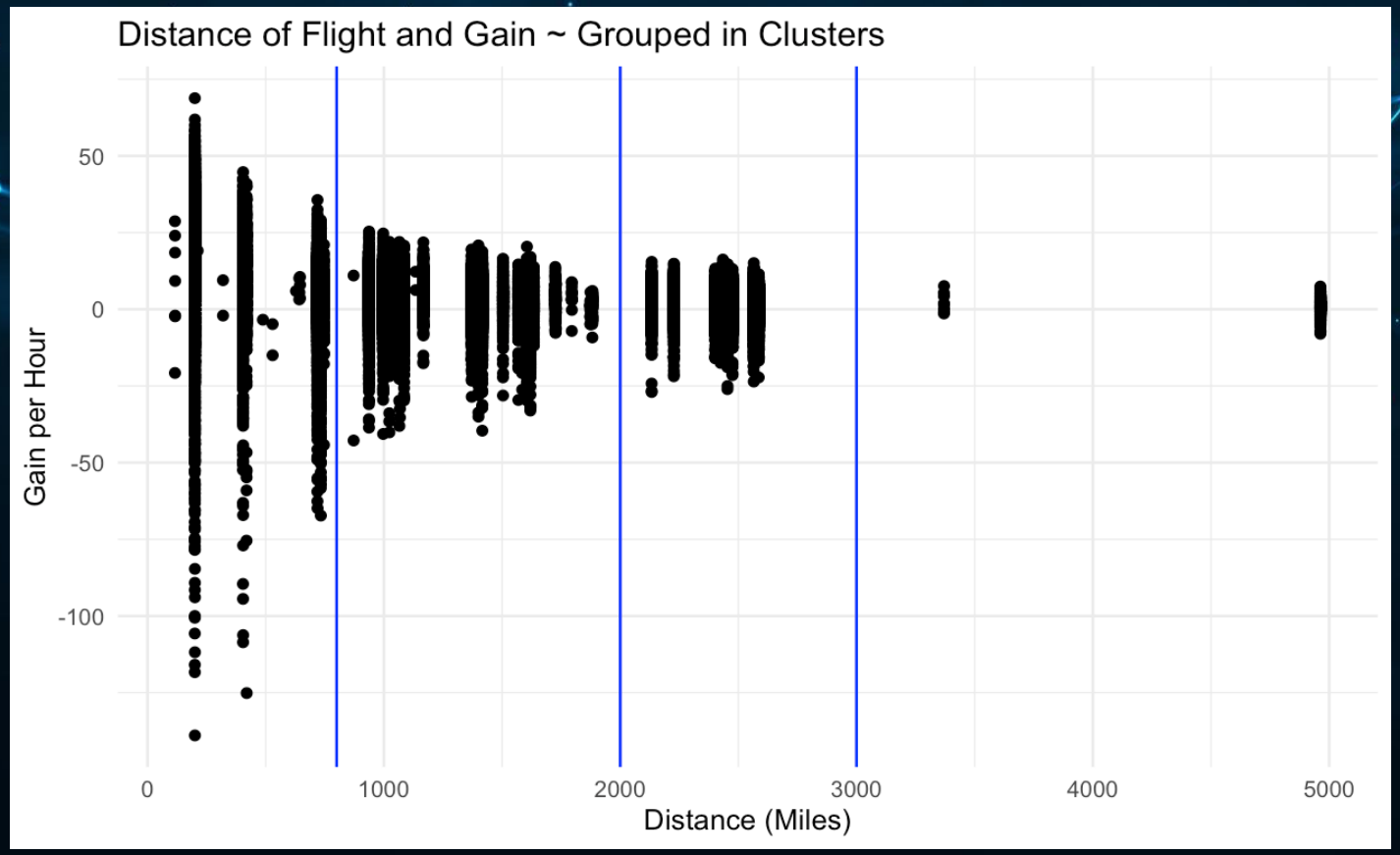
Preventing departure delays matters more than expecting pilots to "make up" time in the air.

Distribution of Gain per Hour by Departure Status



High variability and negative outliers show that in-air time recovery is unpredictable, reinforcing the importance of preventing departure delays.

Do Long Flights Have Different Average Gain per Hour than Short Flights?

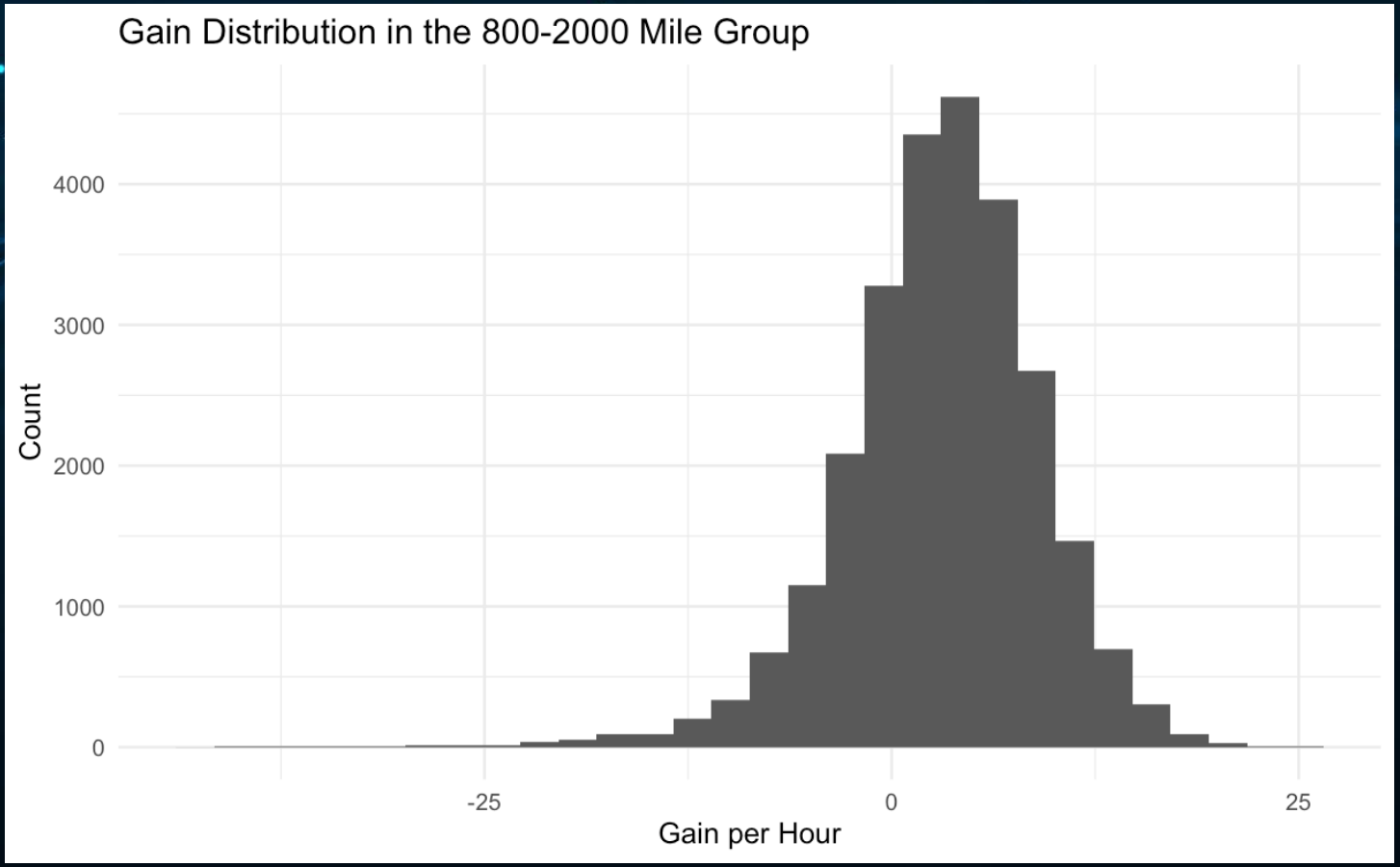


Groups with similar flight distances were created:

- 0-800
- 800-2000
- 2000-3000
- 3000-5000

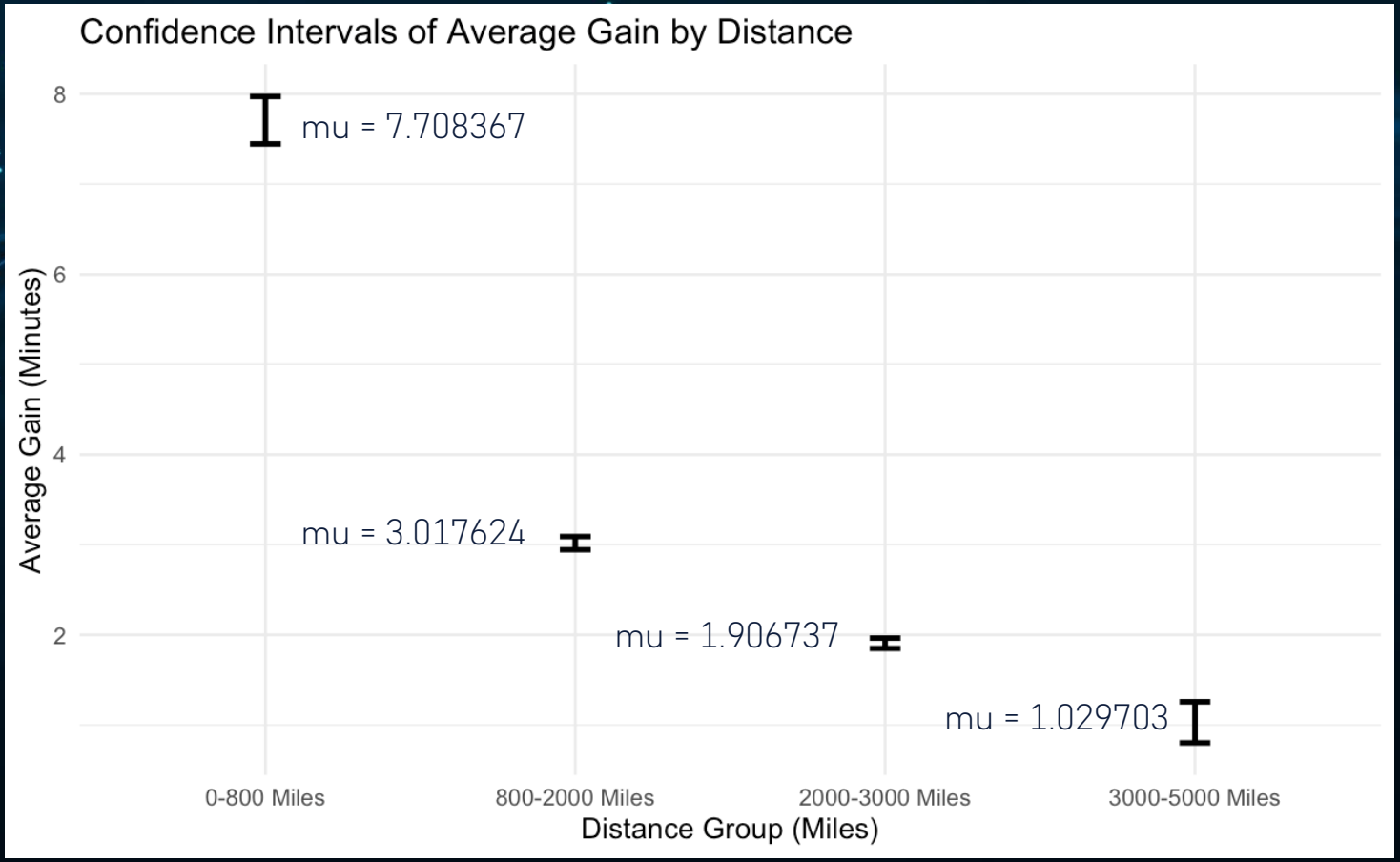
Do Long Flights Have Different Average Gain per Hour than Short Flights?

- To use confidence intervals, normality must be established
- Normality of Gain was estimated using distribution plots for each group
- The graph here shows the plot for the 800-2000 mile group



Do Long Flights Have Different Average Gain per Hour than Short Flights?

- Confidence intervals were created for the average gain of each group
- Longer flights have less gain than short flights, seen in the decreasing trend across each group



Conclusion

- UA flights see meaningful difference in average gain based on delay time.
- UA flights recovered about 7-9 minutes in the air.
- Factors like flight duration and destination play a meaningful role.
- Longer flights had less average gain than shorter flights
- Flights can recover a little time in the air, but it's usually not enough to fix big delays